The documentation and process conversion measures necessary to comply with this revision shall be completed by 4 July 2013.

INCH-POUND

MIL-PRF-19500/744D
4 January 2013
SUPERSEDING
MIL-PRF-19500/744C
1 May 2012

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR, N-CHANNEL,
* RADIATION HARDENED, LOGIC-LEVEL SILICON,
TYPES 2N7616UB, 2N7616UBC, 2N7616UBN, 2N7616UBCN,
JANTXVR, JANTXVF, JANSR, AND JANSF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for a N-channel, enhancement-mode, radiation hardened, low-threshold logic level, MOSFET, transistor. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500, with avalanche energy maximum rating ($E_{AS}$) and maximum avalanche current ($I_{AS}$). See 6.5 for JANHC and JANKC die versions.

1.2 Physical dimensions. See figure 1, UB (metal lid, as shield, connected to fourth pad), UBC (ceramic lid, braze-ring connected to fourth pad), UBN (3-pin, isolated metal lid), and UBCN (3-pin, isolated ceramic lid).

1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C.

| Type | $P_T$ $T_A$ = +25°C (1) | $P_T$ (infinite sink) $T_{IS}$ = +25°C | $R_{\theta JA}$ (2) | $V_{DS}$ | $V_{GS}$ | $I_{D1}$ $T_C$ = +25°C (3) (4) | $I_{D2}$ $T_C$ = +100°C (3) (4) | $I_S$ | $I_{DM}$ (5) | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | V dc | V dc | A dc | A dc | A dc | A (pk) | °C |
| 2N7616UB, UBC, UBN, UBCN | 0.62 | 1.25 | 200 | 60 | ±10 | 0.8 | 0.5 | 0.8 | 3.2 | -55 to +150 |

(1) Derate linearly by 4.5 mW/°C for $T_A$ > +25°C
(2) See figure 2, thermal impedance curves.
(3) The following formula derives the maximum theoretical $I_D$ limit:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(4) See figure 3, maximum drain current graph.
(5) $I_{DM}$ = 4 X $I_{D1}$ as calculated in note (3).

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.4 Maximum ratings. Unless otherwise specified, $T_C$ = +25°C.

| Type | Min $V_{(BR)\ DSS}$ $V_{GS}$ = 0 V $I_D$ = 250 µA dc | $V_{GS\ (th)1}$ $V_{DS} = V_{GS}$ $I_D$ = 250 µA dc | Max $I_{DSS1}$ $V_{GS}$ = 0 V $V_{DS}$ = 80% rated VDS | Max $r_{DS(on)}$ (1) $V_{GS}$ = 4.5 V dc at $I_{D2}$ $T_J$ = +25°C | Max $r_{DS(on)}$ (1) $V_{GS}$ = 4.5 V dc $T_J$ = +150°C | $E_{AS}$ | $I_{AS}$ |
|---|---|---|---|---|---|---|---|
| | V dc | V dc Min | Max | µA dc | Ohm | Ohm | mJ | A dc |
| 2N7616UB UBC, UBN, UBCN | 60 | 1.0 | 2.0 | 1.0 | 0.680 | 1.020 | 26.6 | 1.0 |

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or https://assist.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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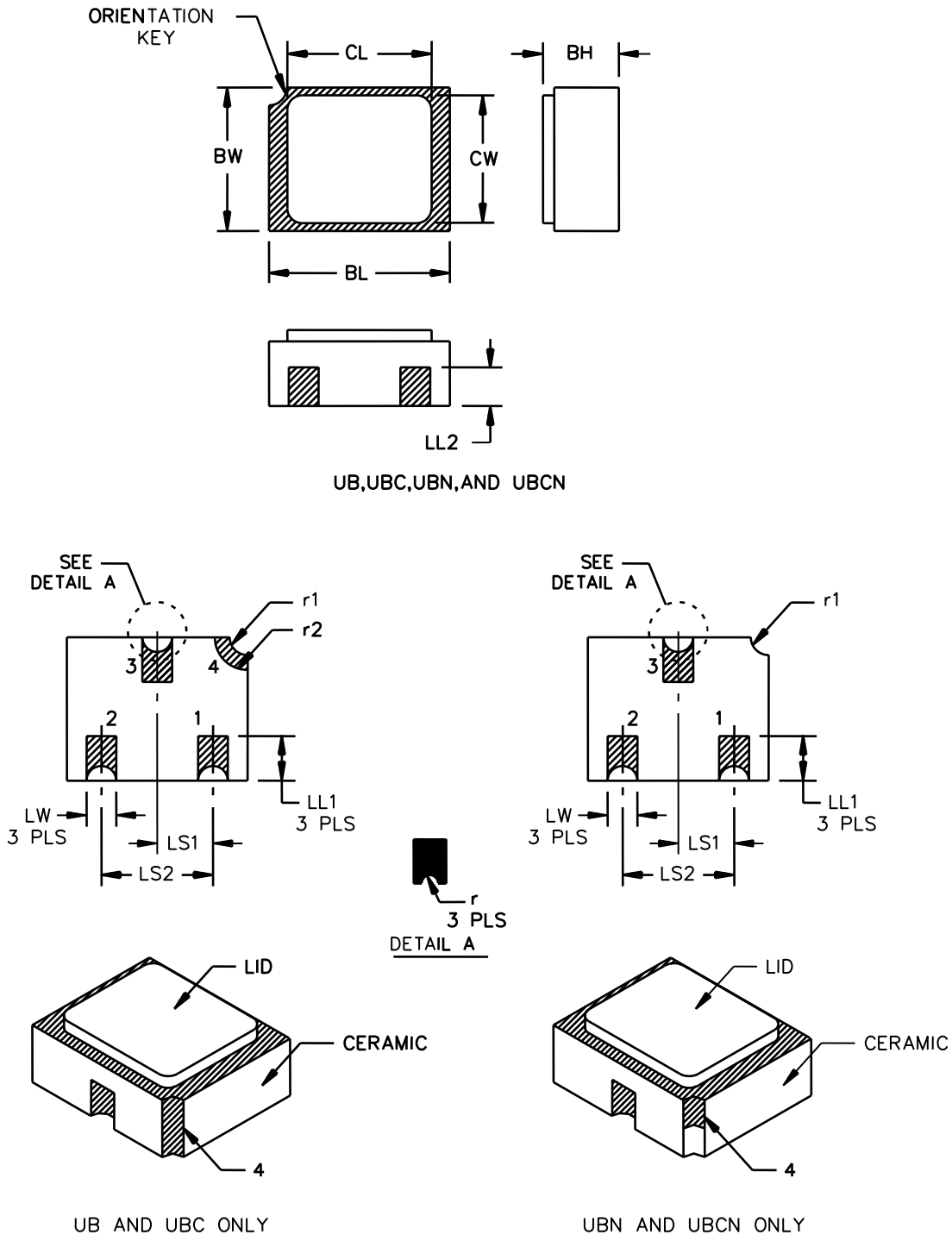



FIGURE 1. Physical dimensions, surface mount (UB, UBN, UBC and UBCN versions).



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| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BL | .115 | .128 | 2.92 | 3.25 | |
| BW | .095 | .108 | 2.41 | 2.74 | |
| BH | .046 | .056 | 1.17 | 1.42 | UB only, 4 |
| BH | .046 | .056 | 1.17 | 1.42 | UBN only, 5 |
| BH | .055 | .069 | 1.40 | 1.75 | UBC only, 6 |
| BH | .055 | .069 | 1.40 | 1.75 | UBCN only, 7 |
| CL | | .128 | | 3.25 | |
| CW | | .108 | | 2.74 | |
| LL1 | .022 | .038 | 0.56 | 0.97 | 3 PLS |
| LL2 | .014 | | 0.356 | | 3 PLS |
| $LS_1$ | .035 | .039 | 0.89 | 0.99 | |
| $LS_2$ | .071 | .079 | 1.80 | 2.01 | |
| LW | .016 | .024 | 0.41 | 0.61 | |
| r | | .008 | | 0.20 | 6 |
| r1 | | .012 | | 0.30 | 8 |
| r2 | | .022 | | 0.56 | UB & UBC only, 8 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. UB only: Pad 1 = Gate, Pad 2 = Source, Pad 3 = Drain, Pad 4 = Shielding connected to the metal lid.
5. UBN only: Pad 1 = Gate, Pad 2 = Source, Pad 3 = Drain, Isolated lid with 3 pads only.
6. UBC (ceramic lid) only: Pad 1 = Gate, Pad 2 = Source, Pad 3 = Drain, Pad 4 = Connected to the lid braze ring.
7. UBCN (ceramic lid) only: Pad 1 = Gate, Pad 2 = Source, Pad 3 = Drain, Isolated lid with 3 pads only.
8. For design reference only.
9. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 1. Physical dimensions, surface mount (UB, UBN, UBC and UBCN versions) - Continued.



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

$I_{AS}$ ..... Rated avalanche current, nonrepetitive
nC ..... nano Coulomb.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 (UB, UBN, UBC and UBCN) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturer’s symbol or logo. The prefixes JANTXV and JANS can be abbreviated as JV and JS respectively. The "2N" prefix and the "UB" suffix can also be omitted. The radiation hardened designator R, F, G, or H shall immediately precede (or replace) the device “2N” identifier (depending upon degree of abbreviation required).

3.6 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.6.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ kΩ, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.



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4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.2.1.1 SEE. SEE shall be performed at initial qualification and after process or design changes which may affect radiation hardness (see table III and table IV). Upon qualification, manufacturers shall provide the verification test conditions from section 5 of method 1080 of MIL-STD-750 that were used to qualify the device for inclusion into section 6 of the slash sheet. End-point measurements shall be in accordance with table II. SEE characterization data shall be made available upon request of the qualifying or acquiring activity.



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4.3 Screening (JANS and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) (1) (2) | Measurement | |
|---|---|---|
| | JANS | JANTXV |
| (3) | Gate stress test (see 4.3.1) | Gate stress test (see 4.3.1) |
| (3) | Method 3470 of MIL-STD-750, $E_{AS}$ (see 4.3.2) | Method 3470 of MIL-STD-750, $E_{AS}$ (see 4.3.2) |
| (3) 3c | Method 3161 of MIL-STD-750, thermal impedance, (see 4.3.3) | Method 3161 of MIL-STD-750, thermal impedance, (see 4.3.3) |
| 9 | Subgroup 2 of table I herein<br>$I_{DSS1}$, $I_{GSSF1}$, $I_{GSSR1}$ as minimum | Not applicable |
| 10 | Method 1042 of MIL-STD-750, test condition B | Method 1042 of MIL-STD-750, test condition B |
| 11 | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(ON)1}$, $V_{GS(TH)1}$<br>Subgroup 2 of table I herein.<br><br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±0.2 µA dc or ±100 percent of initial value, whichever is greater. | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(ON)1}$, $V_{GS(TH)1}$<br>Subgroup 2 of table I herein. |
| 12 | Method 1042 of MIL-STD-750, test condition A | Method 1042 of MIL-STD-750, test condition A |
| 13 | Subgroups 2 and 3 of table I herein<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±0.2 µA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{DS(ON)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(TH)1}$ = ±20 percent of initial value. | Subgroup 2 of table I herein<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±0.2 µA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{DS(ON)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(TH)1}$ = ±20 percent of initial value. |

(1) At the end of the test program, $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ are measured.
(2) An out-of-family program to characterize $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a. JANTXV level does not need to be repeated in screening requirements.



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4.3.1 Gate stress test. Apply $V_{GS}$ = 15 V minimum for t = 250 μs minimum.

4.3.2 Single pulse avalanche energy ($E_{AS}$).

a. Peak current ($I_{AS}$).....1.0 A.

b. Peak gate voltage ($V_{GS}$).....10 V dc (up to max rated $V_{GS}$).

c. Gate to source resistor ($R_{GS}$).....$25 \leq R_{GS} \leq 200\ \Omega$.

d. Initial case temperature .....+25°C, +10°C, -5°C.

e. Inductance: ..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.

f. Number of pulses to be applied .....1 pulse minimum.

g. Supply voltage ($V_{DD}$).....25 V dc (up to max $V_{DS}$).

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, (and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. See table III, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JANTXV) of MIL-PRF-19500, and herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1051 | Test condition G, 100 cycles. |
| B3 | 2077 | SEM. |
| B5 | 1042 | Accelerated steady-state gate bias, condition B, $V_{GS}$ = rated $V_{GS}$; $T_A$ = +175°C, t = 24 hours minimum; or $T_A$ = +150°C, t = 48 hours minimum. |
| B5 | 1042 | Accelerated steady-state reverse bias, condition A, $V_{DS}$ = rated $V_{DS}$; $T_A$ = +175°C, t = 120 hours minimum; or $T_A$ = +150°C, t = 240 hours minimum. |
| B5 | 2037 | Test condition D. |



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4.4.2.2 Group B inspection, table E-VIB (JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 1051 | Test condition C, 25 cycles. |
| B3 | 1042 | Intermittent operation life, condition D, 2,000 cycles. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Terminal strength is not applicable. |
| C5 | 3161 | See 4.3.3, $R_{\theta JA}$ = (see 1.3). |
| C6 | 1042 | Intermittent operation life, condition D, 6,000 cycles. |

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 2/ | 3161 | See 4.3.3 | $Z_{\theta JA}$ | | | °C/W |
| Breakdown voltage drain to source | 3407 | $V_{GS}$ = 0, $I_D$ = 250 µA dc, bias condition C | $V_{(BR)DSS}$ | 60 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 250 µA dc | $V_{GS(TH)1}$ | 1.0 | 2.0 | V dc |
| Gate current | 3411 | $V_{GS}$ = +10 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSSF1}$ | | +100 | nA dc |
| Gate current | 3411 | $V_{GS}$ = -10 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSSR1}$ | | -100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0, bias condition C, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS1}$ | | 1.0 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 4.5 V dc, condition A, pulsed (see 4.5.1), $I_D$ = $I_{D2}$ | $r_{DS(ON)1}$ | | 0.680 | Ω |
| Forward voltage | 4011 | $V_{GS}$ = 0, condition A, pulsed (see 4.5.1), $I_D$ = $I_{D1}$ | $V_{SD}$ | | 1.2 | V (pk) |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_C$ = $T_J$ = +125°C | | | | |
| Gate current | 3411 | $V_{GS}$ = ±10 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSS2}$ | | ±200 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0, bias condition C, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS2}$ | | 10 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 4.5 V dc, condition A, pulsed (see 4.5.1), $I_D$ = $I_{D2}$ | $r_{DS(ON)2}$ | | 0.980 | Ω |
| Gate to source voltage (threshold) | 3403 | $V_{DS}$ = $V_{GS}$, $I_D$ = 250 µA dc | $V_{GS(TH)2}$ | 0.5 | | V dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 3 – Continued. | | | | | | |
| Low temperature operation: | | $T_C = T_J = -55°C$ | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 250 µA dc | $V_{GS(TH)3}$ | | 2.5 | V dc |
| Subgroup 4 | | | | | | |
| Forward transconductance | 3475 | $V_{DS}$ = 10 V dc, $I_D = I_{D2}$, pulsed (see 4.5.1) | $g_{FS}$ | 0.23 | | S |
| Gate series resistance | 3402 | Condition B | $R_G$ | | 14 | Ω |
| Subgroup 5 | | | | | | |
| Safe operating area test | 3474 | $V_{DS}$ = 80 percent of rated $V_{DS}$ (see 1.3); $t_P$ = 10 ms, $I_D$ as specified on figure 4 | | | | |
| Electrical measurements | | See table I, subgroup 2 | | | | |
| Subgroups 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | 3471 | Condition B, $I_D = I_{D1}$, $V_{DD}$ = 50 percent rated $V_{DS}$ | | | | |
| On-state gate charge | | | $Q_{G(on)}$ | | 3.6 | nC |
| On gate to source charge | | | $Q_{GS1}$ | | 1.5 | nC |
| On gate to drain charge | | | $Q_{GD1}$ | | 1.8 | nC |
| Turn-off gate charge | | | $Q_{G(off)}$ | | 3.6 | nC |
| Off gate to source charge | | | $Q_{GS2}$ | | 1.5 | nC |
| Off gate to drain charge | | | $Q_{GD2}$ | | 1.8 | nC |
| Reverse recovery time | 3473 | di/dt = -100 A/µs, $V_{DD} \leq$ 25 V, $I_D = I_{D1}$ | $t_{rr}$ | | 78 | ns |

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 3 and 4 (JANS).
Group B, subgroups 2 and 3 (JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.



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TABLE II. Group D inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | Post-irradiation limits | | Post-irradiation limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | R and F | | R | | F | | |
| | Method | Conditions | | Min | Max | Min | Max | Min | Max | |
| Subgroup 1 | | | | | | | | | | |
| Not applicable | | | | | | | | | | |
| Subgroup 2 | | $T_C$ = + 25°C | | | | | | | | |
| Steady-state total dose irradiation ($V_{GS}$ bias) 4/ | 1019 | $V_{GS}$ = 10 V; $V_{DS}$ = 0 | | | | | | | | |
| Steady-state total dose irradiation ($V_{DS}$ bias) 4/ | 1019 | $V_{GS}$ = 0; $V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | | | | | | | | |
| End-point electricals: | | | | | | | | | | |
| Breakdown voltage, drain to source | 3407 | $V_{GS}$ = 0; $I_D$ = 250 µA; bias condition C | $V_{(BR)DSS}$ | 60 | | 60 | | 60 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$ $I_D$ = 250 µA | $V_{GS(th)1}$ | 1.0 | 2.0 | 1.0 | 2.0 | 1.0 | 2.0 | V dc |
| Gate current | 3411 | $V_{GS}$ = +10 V, $V_{DS}$ = 0, bias condition C | $I_{GSSF1}$ | | 100 | | 100 | | 100 | nA dc |
| Gate current | 3411 | $V_{GS}$ = -10 V, $V_{DS}$ = 0, bias condition C | $I_{GSSR1}$ | | -100 | | -100 | | -100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0, bias condition C; $V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | $I_{DSS}$ | | 1.0 | | 1.0 | | 1.0 | µA dc |
| Static drain to source on-state voltage | 3405 | $V_{GS}$ = 4.5 V; condition A, pulsed (see 4.5.1), $I_{D1} = I_{D2}$ | $V_{DS(on)}$ | | 0.275 | | 0.275 | | 0.275 | V dc |
| Forward voltage source drain diode | 4011 | $V_{GS}$ = 0; $I_D$ = $I_{D1}$ bias condition C | $V_{SD}$ | | 1.2 | | 1.2 | | 1.2 | V dc |

1/ For sampling plan see MIL-PRF-19500.
2/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification sheets utilizing the same die design.
3/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in its qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.
4/ Separate samples shall be pulled for each bias.



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* TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices c = 0 |
| Temperature cycling | 1051 | Condition G, 500 cycles | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 2 1/ | | | 45 devices c = 0 |
| Steady-state gate bias | 1042 | Condition B, 1,000 hours | |
| Electrical measurements | | See table I, subgroup 2 | |
| Steady-state reverse bias | 1042 | Condition A, 1,000 hours | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 3 | | | 45 devices c = 0 |
| Switching time test | 3472 | $I_D = I_{D1}$ , $V_{GS}$= 5.0 V dc, $R_G$ = 24Ω,<br>$V_{DD}$ = 50 percent rated $V_{DS}$<br>Maximum measurements: $t_{d(on)}$ = 8 ns; $t_r$ = 24 ns;<br>$t_{d(off)}$ = 30 ns; $t_f$ = 13 ns | |
| Subgroup 4 | | | Sample size N/A |
| Thermal impedance curves | 3161 | See MIL-PRF-19500. | |
| Subgroup 10 | | | 22 devices c = 0 |
| Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors | 3476 | Test conditions shall be derived by the manufacturer. | |
| Subgroup 11 | | | 3 devices |
| SEE 2/ 3/ | 1080 | See MIL-STD-750 method 1080 and 6.2. | |

1/ A separate sample for each test shall be pulled.
2/ Group E qualification of SEE effect testing may be performed prior to lot formation. Qualification may be extended to other specification sheets utilizing the same structurally identical die design.
3/ Device qualification to a higher level LET is sufficient to qualify all lower level LETs.



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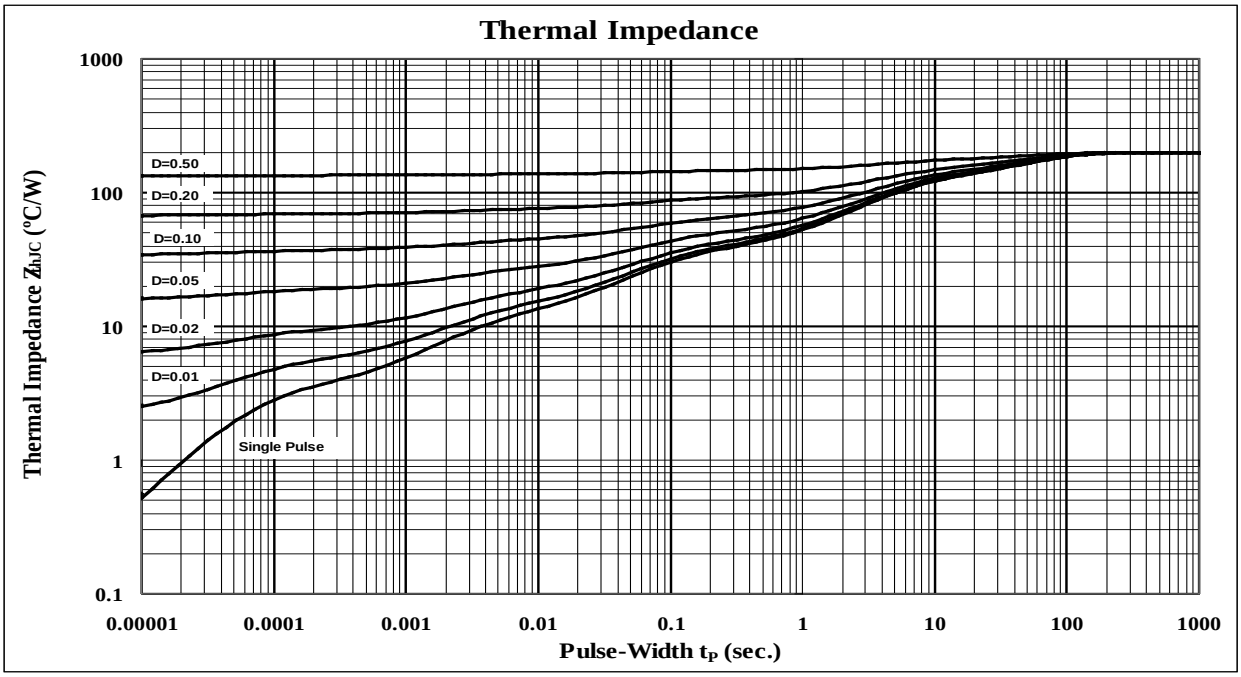



FIGURE 2. Thermal impedance graph.



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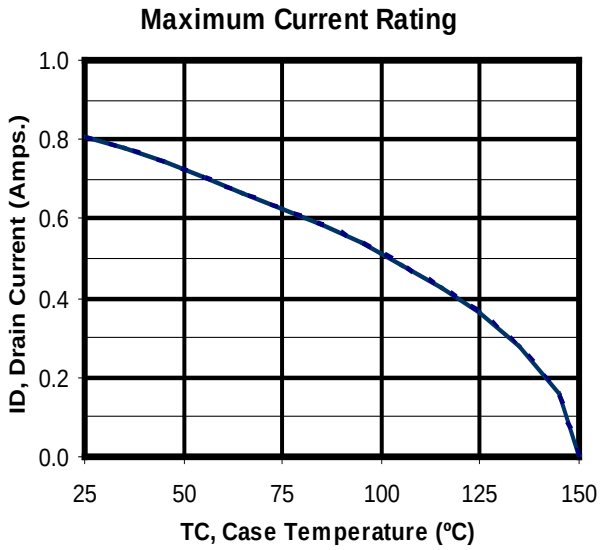



FIGURE 3. Derating drain current .



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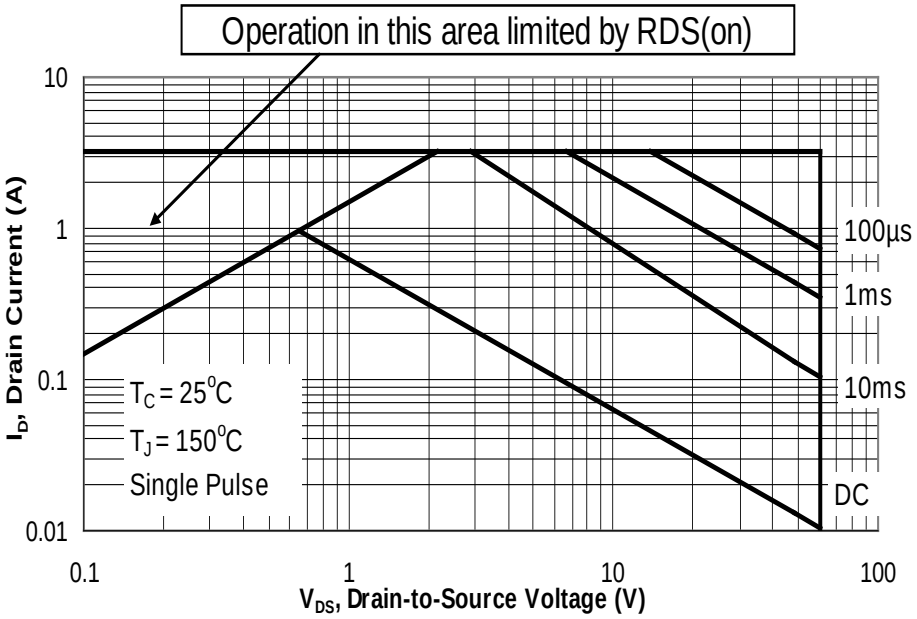



FIGURE 4. Safe-operating-area graph.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it should be specified in the contract or order.

* f. If specific SEE characterization conditions are desired (see section 6.6 and table IV), manufacturer's cage code should be specified in the contract or order.

* g. If SEE testing data is desired, it should be specified in the contract or order.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 JANC die versions. The JANHC and JANKC die versions of these devices are covered under specification sheet MIL-PRF-19500/741.



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6.5 Cross-reference list. The following table shows the generic P/N and its associated military P/N (without JAN and RHA prefix).

| Generic P/N | Military P/N | Package & Termination Configuration |
|---|---|---|
| IRHLUB770Z4 | 2N7616UB | 4-pad, Metal Lid (Shield) connected to 4th pad |
| IRHLUBN770Z4 | 2N7616UBN | 3-pad, Isolated Metal Lid |
| IRHLUBC770Z4 | 2N7616UBC | 4-pad, Ceramic Lid, lid braze ring connected to 4th pad |
| IRHLUBCN770Z4 | 2N7616UBCN | 3-pad, Isolated Ceramic Lid |

* 6.6 Application data.

* 6.6.1 Manufacturer specific irradiation data. Each manufacturer qualified to this slash sheet has characterized its devices to the requirements of MIL-STD-750 method 1080 and as specified herein. Since each manufacturer's characterization conditions can be different and can vary by the version of method 1080 qualified to, the MIL-STD-750 method 1080 revision version date and conditions used by each manufacturer for characterization have been listed here (see table IV) for information only. SEE conditions and figures listed in section 6 are current as of the date of this specification sheet, please contact the manufacturer for the most recent conditions.



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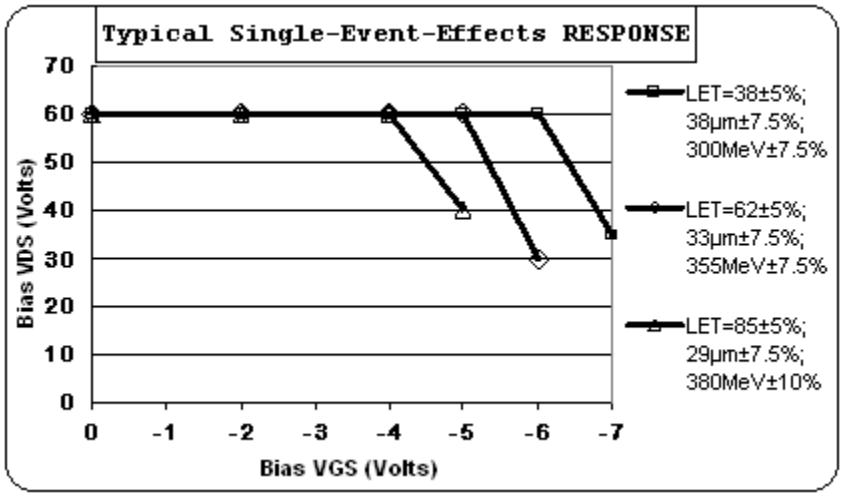
* TABLE IV. Manufacturers characterization conditions.

| Manufacture s cage | Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|---|
| | | Method | Conditions | |
| 69210 (Applicable to devices with a date code of 21 August 2012 and older) | SEE 1/ | 1080 | See MIL-STD-750E method 1080.0 dated 20 November 2006. See figure 5 | |
| | Electrical measurements | | $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ in accordance with table I, subgroup 2 | 3 devices |
| | SEE irradiation: | | Fluence = 3E5 ±20 percent ions/cm$^2$<br>Flux = 2E3 to 2E4 ions/cm$^2$/sec, temperature = 25º ±5 °C | |
| | | | Surface LET = 38 MeV-cm$^2$/mg ±5.0 %, range = 38 µm ±7.5%,<br>energy = 300 MeV ±7.5%<br>(nominal 3.86 MeV/nucleon at Brookhaven National Lab Accelerator)<br>In-situ bias conditions: $V_{DS}$ = 60 V and $V_{GS}$ = -6 V<br>$V_{DS}$ = 35 V and $V_{GS}$ = -7 V | |
| | | | Surface LET = 62 MeV-cm$^2$/mg ±5.0 %, range = 33 µm ±7.5%,<br>energy = 355 MeV ±7.5%<br>(nominal 2.92 MeV/nucleon at Brookhaven National Lab Accelerator)<br>In-situ bias conditions: $V_{DS}$ = 60 V and $V_{GS}$ = -5 V<br>$V_{DS}$ = 30 V and $V_{GS}$ = -6 V | |
| | | | Surface LET = 85 MeV-cm$^2$/mg ±5 %, range = 29 µm ±7.5%,<br>energy = 380 MeV ±10%<br>(nominal 1.98 MeV/nucleon at Brookhaven National Lab Accelerator)<br>In-situ bias conditions: $V_{DS}$ = 60 V and $V_{GS}$ = -4 V<br>$V_{DS}$ = 40 V and $V_{GS}$ = -5 V | |
| | Electrical measurements | | $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ in accordance with table I, subgroup 2 | |
| Upon qualification, all manufacturers shall provide the verification test conditions to be added to this table. | | | | |

1/ $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ was examined before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions in accordance with table I, subgroup 2, may be performed at the manufacturer's option.



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* FIGURE 5. Cage 68210 typical SEE response graph.



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6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2012-076)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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